Joseph Telaak

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Education

University of South Carolina Columbia, SC
BSE in Computer Engineering, Leadership Distinction in Research (GPA: 3.61) *Aug. 2022 – Dec. 2024*

Experience

Chief Executive Officer Dec. 2024 - Present
Parkeze Inc. *Columbia, SC*

- Ported the LoRaWAN gateway stack to Zephyr RTOS with custom high-availability features.
- Developed sensor firmware integrating magnetometer and mmWave radar for vehicle occupancy detection.
- Architected cloud-native backend with Kafka ingestion pipeline supporting real-time streaming to 1000+ users.
- Built analytics platform using React, GraphQL, and TimescaleDB with custom Keycloak IAM extensions.
- Created CI/CD pipeline for integration testing and blue/green deployment on Kubernetes.
- Involved in all aspects of the business including fundraising, sales, and customer support.

Research Assistant Feb. 2023 - Dec. 2024
USC Systems Research on X Lab *Columbia, SC*

- Developed firmware for live data streaming from TI IWR series 77–81 GHz mmWave radar SoCs.
- Implemented contactless vital sign monitoring via chest-wall vibration analysis, validated against ECG.
- Co-authored research on novel mmWave radar algorithms for 3D vehicle and pedestrian detection.
- Designed FPGA-based data collection infrastructure for multi-radar synchronization and high-throughput capture.
- Secured research funding through competitive grant proposals.

Project Lead Jan. 2022 - Feb. 2023
SCGSSM Autonomous Golf Cart *Hartsville, SC*

- Engineered a complete ADAS control module for steering, throttle, and braking actuation.
- Designed custom Nvidia Jetson carrier board with multi-channel analog video capture and ADC interfacing.
- Implemented real-time computer vision pipeline for obstacle detection, sign recognition, and lane tracking.
- Integrated CAN bus and low-level actuator control for drive-by-wire operation of the vehicle platform.

Leadership

FIRST Robotics Columbia, SC
FIRST Technical Advisor, Judge, Robot Inspector *Jan. 2022 - May 2025*

Selected Projects

LLM Voice Assistant: Far-field mic array with XMOS DSP, Zephyr RTOS on i.MXRT, ESP32 Wi-Fi NIC, multi-chip OTA, and real-time audio streaming to an LLM inference pipeline.

Agentic Audio Streamer: Multi-room audio system using SigmaDSP, DACs, and Raspberry Pi CM5 with Spotify Connect and a custom LLM control API.

PiSwitch: 7-port router/switch with custom ASIC controller, web UI, and out-of-band management interface.

RISCv CPU: Pipelined RISC-V CPU with GPIO peripheral support implemented on FPGA in SystemVerilog.

Pick-n-Place Machine: Designed and built a desktop PCB assembly machine running Marlin firmware and OpenPNP vision software.

Skills

Languages: C/C++, Python, Rust, Java, JavaScript/TypeScript, MATLAB, LUA
Embedded: Zephyr RTOS, Yocto, Buildroot, FreeRTOS, LoRaWAN, I2C, SPI, UART, CAN, USB
Hardware: Altium, KiCad, Quartus, STM32 Cube, MCUXpresso, mmWave Studio, JTAG, oscilloscope
Systems: Linux, Kubernetes, Docker, Kafka, GraphQL, PostgreSQL, CI/CD, Git
Memberships: IEEE Eta Kappa Nu (HKN), IEEE MTTS, IEEE, ACM, AIAA